PYTHON

Python for Data Analysts

Only the Pandas you need, a 30-day plan from zero, and 10 interview questions.

The 12 Pandas functions

```
import pandas as pd
df = pd.read_csv("sales.csv")                 # 1 load
df.head(); df.info(); df.describe()          # 2 inspect
df.isna().sum(); df.fillna(0); df.dropna()   # 3 missing values
df.drop_duplicates(subset="order_id")        # 4 duplicates
df.loc[df["region"] == "North", ["amount"]]  # 5 select
df.query("amount > 5000 and region == 'North'")  # 6 filter
df.groupby("region").agg(total=("amount","sum"), orders=("order_id","count"))  # 7
pd.merge(orders, customers, on="customer_id", how="left")   # 8 join
df.pivot_table(index="region", columns="month", values="amount", aggfunc="sum")  # 9
df["date"] = pd.to_datetime(df["date"]); df["month"] = df["date"].dt.month  # 10
df["region"].value_counts(normalize=True)    # 11 share
df["tax"] = df["amount"] * 0.18              # 12 vectorise, avoid loops
```

SQL to Pandas

SQL	Pandas
WHERE	`df[df.col > 5]` or `query`
GROUP BY	`groupby().agg()`
JOIN	`merge`
UNION ALL	`pd.concat`
ORDER BY	`sort_values`
ROW_NUMBER	`groupby().cumcount()+1`

30-day plan from zero

Days	Focus
1-5	Variables, lists, dicts, loops, functions
6-12	Pandas: load, clean, filter, groupby, merge
13-17	Dates, string methods, pivot tables
18-22	Charts with Matplotlib and Seaborn
23-27	One end-to-end EDA project on a real dataset
28-30	Push to GitHub, write the README, practise questions

10 interview questions

1. List vs tuple vs dict vs set.
2. loc vs iloc.

3. merge vs join vs concat.
4. How do you handle missing values?
5. apply vs vectorised operations.
6. Find the top 3 products per category.
7. Remove outliers using IQR.
8. Convert a wide table to long (melt).
9. Read a 5 GB CSV that does not fit in memory.
10. Write a function that returns month on month growth.

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